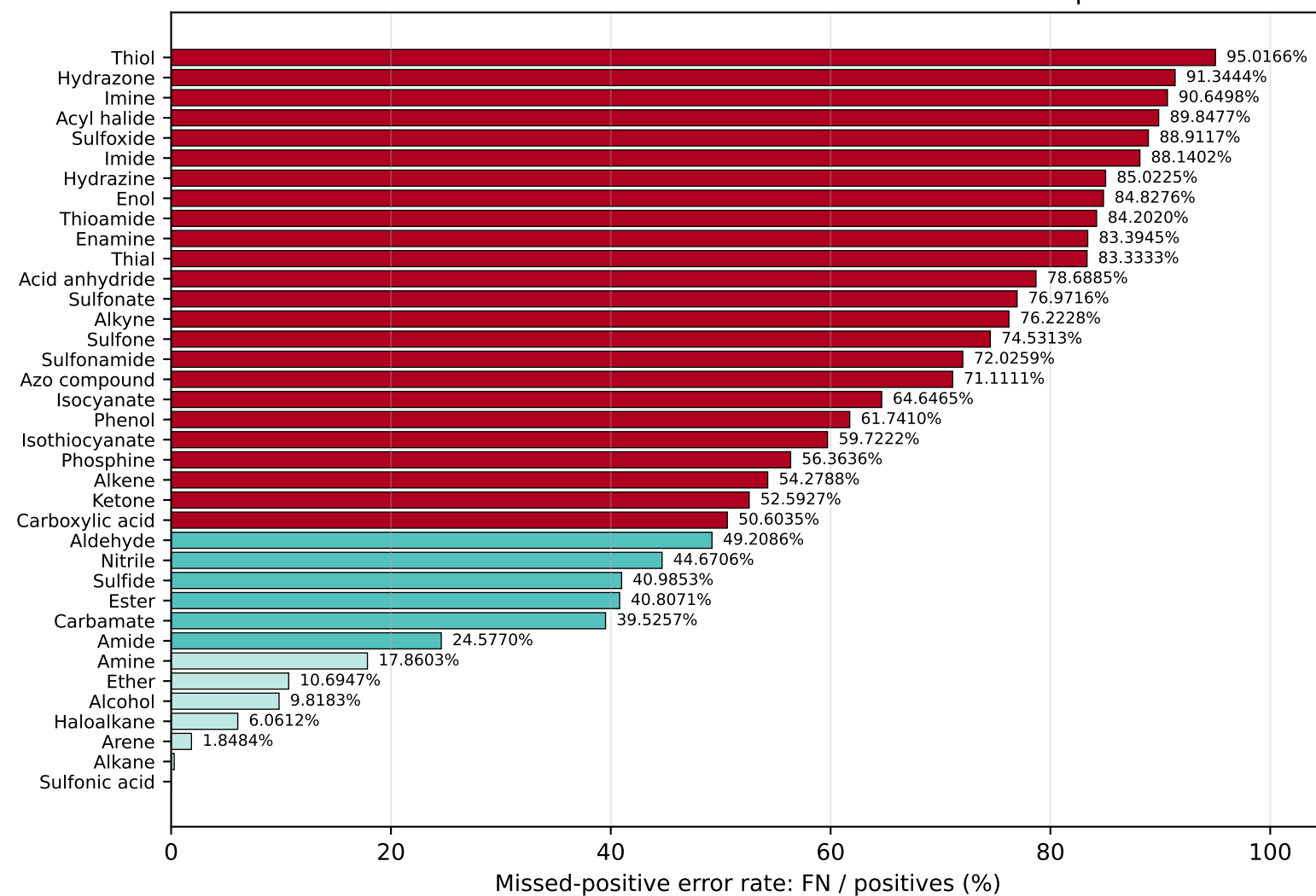
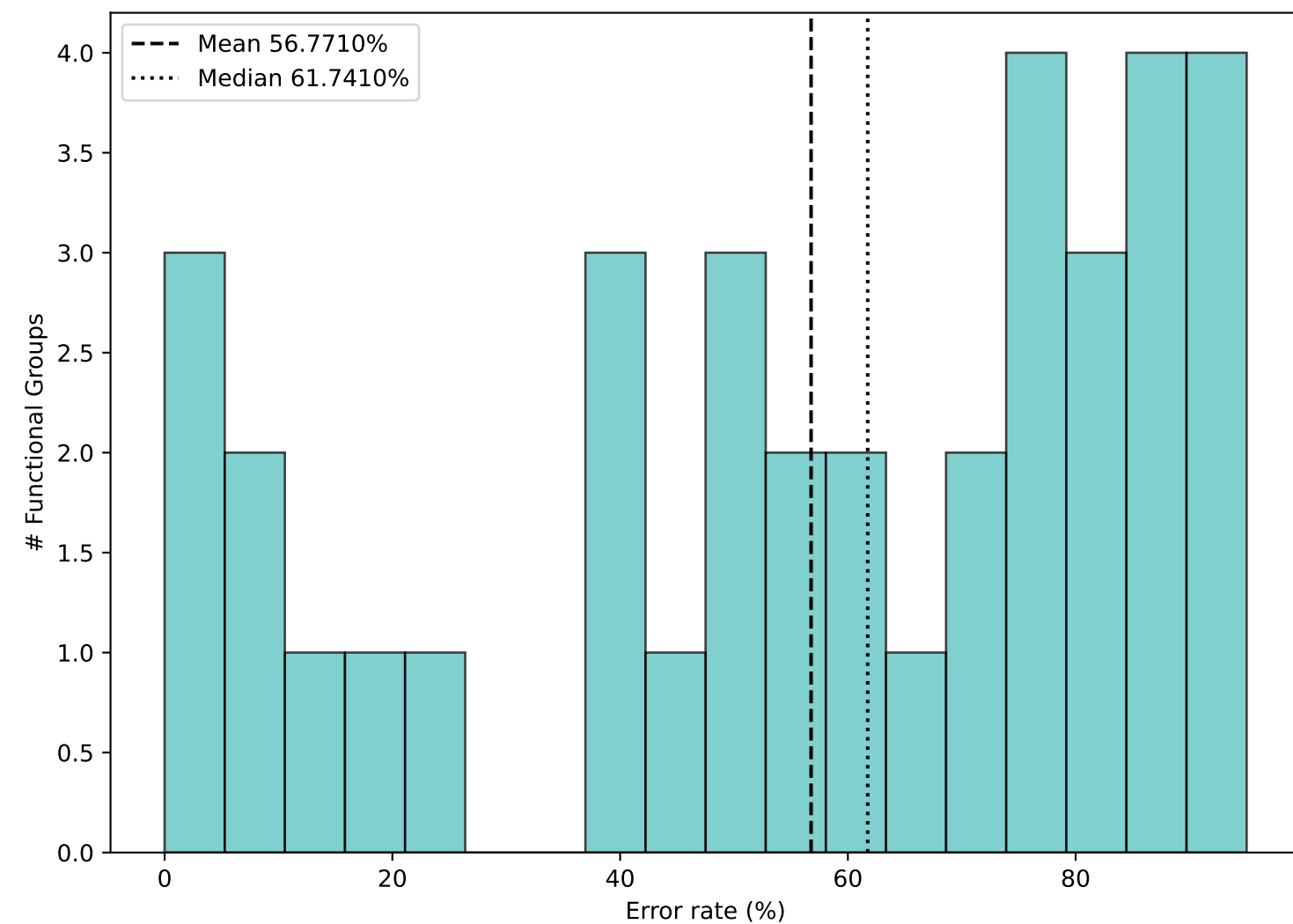


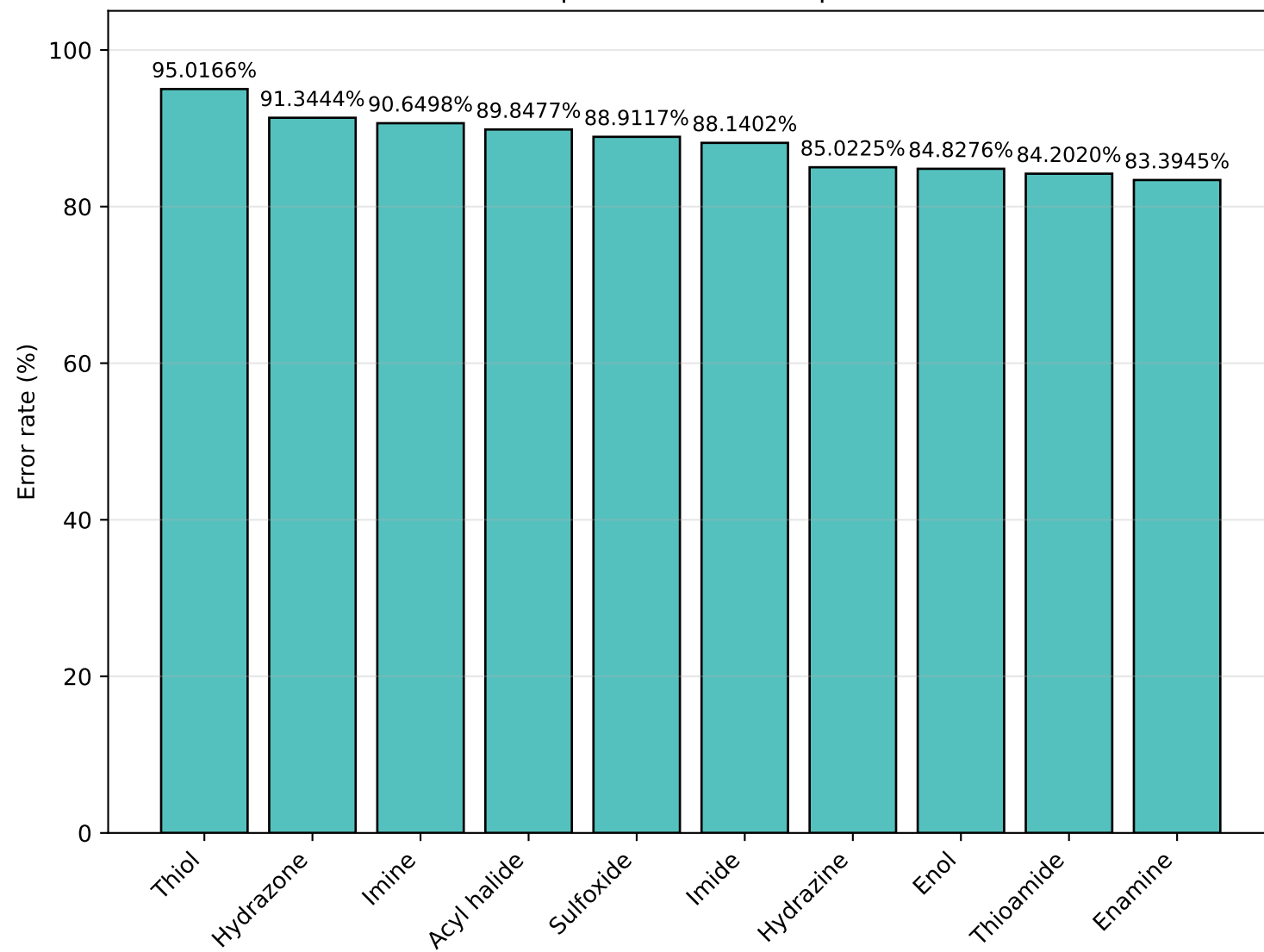
MPS H-NMR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



MPS H-NMR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.6037
F1 macro: 0.2576
Precision micro: 0.4757
Recall micro: 0.8261
Hamming accuracy: 0.8599

Error-rate statistics:

Mean: 56.7710%
Median: 61.7410%
Std: 30.1535%
Max: 95.0166% (Thiol)
Min: 0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%): 24 groups
High (20-50%): 6 groups
Moderate (5-20%): 4 groups
Low (<=5%): 3 groups